

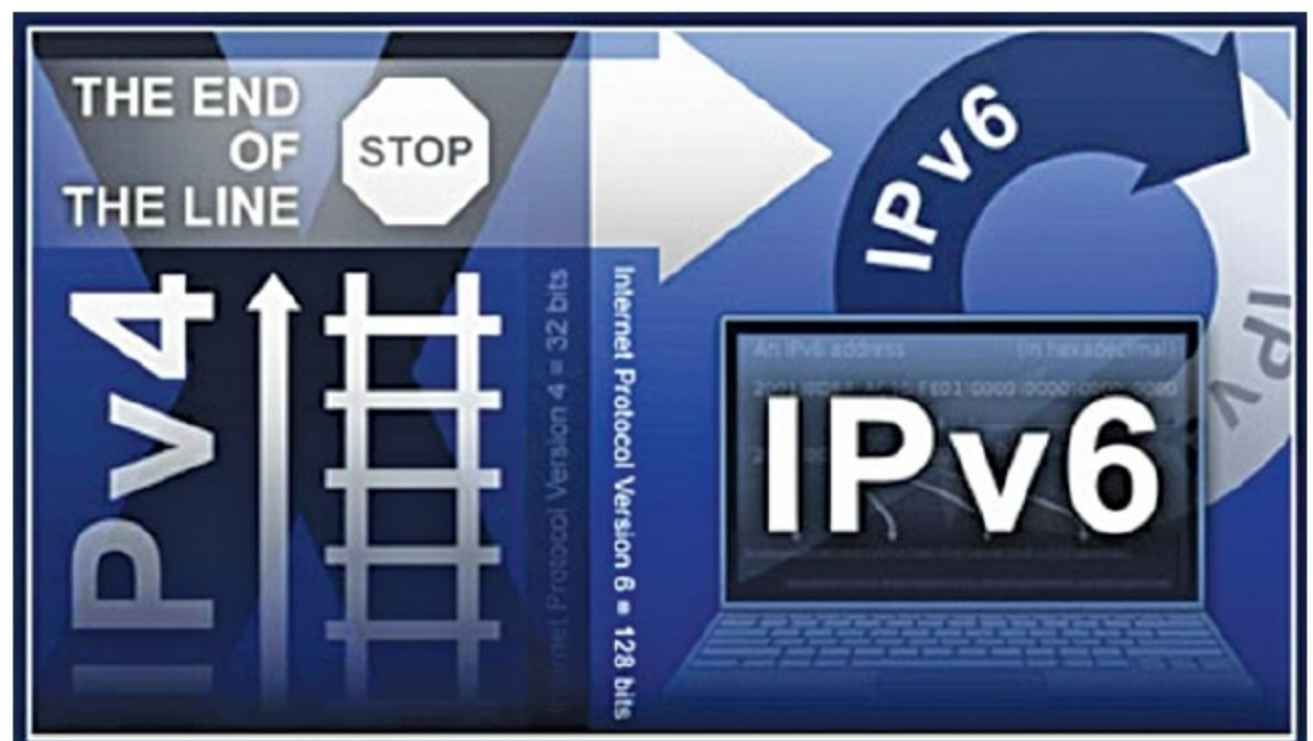
COMMUNICATION

And INTERNET TECHNOLOGY: From IPv4 To IPv6

IPv6 can be compared with IPv4 in terms of two important parameters, namely, address space and IP header simplification. A few people projected the death of the Internet due to increased traffic, yet with IPv6, Internet access is guaranteed, with exponential increase in Internet users over time with the provision of abundant address space



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Let us begin with a simple question: Why go for IP version 6 (IPv6), when IPv4 is already standardised and is widely used across the globe? The answer is: IPv4 is crippled with challenges. First, it has 32-bit address scheme. Due to an exponential growth of Internet users, literally the address space of IPv4 is running out of addresses. The exponential growth rate of Internet users in the world just in the last five years is given below, as per the

statistics of Internet World Statistics (IWS)/International Telecommunication Union (ITU):

- In December 2013, total users = 2802 million, percentage of world population = 39
- In December 2014, total users = 3079 million, percentage of world population = 42.4
- In December 2015, total users = 3366 million, percentage of world population = 46.4